Supplementary Information

A Multi-Track Simulation of Laser-Based Powder Bed Fusion of Metals (PBF-LB/M) with Novel Double Conical Heat Source to Predict the Effects of Laser Turnaround Time on Pad Dimensions and Cooling Rates for NIST AM-Bench 2025 Challenge

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Overview

This Supplementary Information document describes the electronic supplementary materials accompanying the manuscript. The supplementary package contains 23 files: seven transient simulation videos, seven temperature-history videos, eight spreadsheet files containing supporting numerical data, and one image illustrating the steady-state condition. In the manuscript and the journal submission system, these materials are cited consecutively as *Online Resource 1* through *Online Resource 23*.

The supplementary materials provide additional visualization and numerical data supporting the model development, calibration, validation, and analysis presented in the manuscript. Detailed descriptions of the individual online resources are provided in the following sections.

S1. Transient Simulation Videos

Online Resources 1–7 provide transient contour animations of the numerical simulations presented in the manuscript. Each video includes contours of temperature, liquid fraction, volumetric laser heat-source intensity, radiative heat-loss source term, and convective heat-loss source term. Temperature is reported in K, liquid fraction is dimensionless, and the volumetric heat-source intensity and heat-loss source terms are reported in W/m^3.

For each field, contours are presented in the x–z plane, representing a longitudinal section along the laser scanning direction, and in the x–y plane, representing the top view of the build surface. The videos illustrate the temporal evolution of the thermal field, melt-pool formation and solidification, applied volumetric heat source, and heat losses during laser scanning. The opening screen of each video provides the corresponding simulation conditions and numerical parameters, including mesh information, time-step size, track length, laser power, scan speed, and other relevant case-specific parameters.

Online Resource 1 (ESM_1.mp4).
Transient thermal simulation of the NIST AM-Bench 2022 single-track Case 0 using the conventional single-conical volumetric heat-source model. This simulation is used in the single-track validation presented in the main manuscript and serves as the reference heat-source model for comparison with the proposed Double Conical Heat Source (DCHS). The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, and a track length of 2 mm. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 2 (ESM_2.mp4).
Transient thermal simulation of the NIST AM-Bench 2022 single-track Case 0 using the proposed Double Conical Heat Source (DCHS) model. This simulation is used for the single-track validation of the proposed model in the main manuscript and is directly compared with the single-conical heat-source simulation provided in Online Resource 1. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms. The simulation uses the same process conditions as Online Resource 1, with a laser power of 285 W, a scan speed of 960 mm/s, and a track length of 2 mm. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 3 (ESM_3.mp4).
Transient thermal simulation of the NIST AM-Bench 2022 XPad multi-track benchmark using the conventional single-conical volumetric heat-source model. This simulation is used in the multi-track XPad validation presented in the main manuscript and serves as the reference case for comparison with the proposed Double Conical Heat Source (DCHS) model. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during the simulation of 10 bidirectional tracks. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 2 mm, and a hatch spacing of 110 μm. Laser turnaround times of 5 ms and 40 ms were applied after odd- and even-numbered tracks, respectively. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 4 (ESM_4.mp4).
Transient thermal simulation of the NIST AM-Bench 2022 XPad multi-track benchmark using the proposed Double Conical Heat Source (DCHS) model. This simulation is used in the multi-track XPad validation of the proposed model presented in the main manuscript and is directly compared with the conventional single-conical heat-source simulation provided in Online Resource 3. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during the simulation of 10 bidirectional tracks. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 2 mm, and a hatch

spacing of 110 μm. Laser turnaround times of 5 ms and 40 ms were applied after odd- and even-numbered tracks, respectively. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 5 (ESM_5.mp4).
Transient thermal simulation of the NIST AM-Bench 2025 single-track case using the proposed Double Conical Heat Source (DCHS) model. This single-track case was provided as part of the AM-Bench 2025 competition and was used to calibrate the DCHS model prior to its application to the multi-track Challenge 7 cases presented in the main manuscript. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during laser scanning. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 2 mm, and a laser spot diameter of 72 μm. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 6 (ESM_6.mp4).
Transient thermal simulation of the NIST AM-Bench 2025 Challenge 7 multi-track case with a laser turnaround time of 5 ms using the proposed Double Conical Heat Source (DCHS) model. This case constitutes one of the primary multi-track prediction cases of the AM-Bench 2025 competition and was simulated using the DCHS model calibrated from the corresponding single-track case provided in Online Resource 5. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during 15 bidirectional tracks. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 1 mm, a hatch spacing of 110 μm, and a laser spot diameter of 72 μm. A turnaround time of 5 ms was applied between consecutive tracks. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 7 (ESM_7.mp4).
Transient thermal simulation of the NIST AM-Bench 2025 Challenge 7 multi-track case with a laser turnaround time of 0.75 ms using the proposed Double Conical Heat Source (DCHS) model. This case constitutes the second primary multi-track prediction case of the AM-Bench 2025 competition and was simulated using the DCHS model calibrated from the corresponding single-track case provided in Online Resource 5. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during 15 bidirectional tracks. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 1 mm, a hatch spacing of 110 μm, and a laser spot diameter of 72 μm. A turnaround time of 0.75 ms was applied between consecutive tracks. Detailed computational and processing parameters are provided on the opening screen of the video.

S2. Temperature-History Videos

Online Resources 8–14 present the transient temperature histories extracted from the corresponding single-track and multi-track simulations. These videos illustrate the heating and cooling cycles experienced at the selected monitoring locations as the laser passes over or near each point. The temperature histories are used to evaluate the thermal metrics reported in the main manuscript, including the Solid Cooling Rate (SCR), Transition Cooling Rate (TCR), Liquid Cooling Rate (LCR), and Time Above Melt (TAM) for the single-track cases, as well as the Pad Solid Cooling Rate (PSCR) and Pad Time Above Melt (PTAM) for the multi-track cases.

The horizontal threshold lines shown in the temperature-history plots indicate the temperature ranges used to evaluate the corresponding thermal metrics. Temperature is reported in K and time in seconds. The animations show the evolution of the temperature history as the transient simulation progresses and provide a visual representation of the thermal cycles used for the quantitative analyses presented in the main manuscript.

Online Resource 8 (ESM_8.mp4).
Temperature-history animation for the NIST AM-Bench 2022 single-track Case 0 simulated using the conventional single-conical heat-source model. The temperature history corresponds to the single-track validation analysis presented in the main manuscript and is used to evaluate the associated cooling-rate and time-above-melt metrics.

Online Resource 9 (ESM_9.mp4).
Temperature-history animation for the NIST AM-Bench 2022 single-track Case 0 simulated using the proposed Double Conical Heat Source (DCHS) model. The temperature history corresponds to the DCHS single-track validation presented in the main manuscript and is used to evaluate the associated Solid Cooling Rate (SCR), Transition Cooling Rate (TCR), Liquid Cooling Rate (LCR), and Time Above Melt (TAM). This resource provides the thermal-history counterpart to the transient simulation presented in Online Resource 2.

Online Resource 10 (ESM_10.mp4).
Temperature-history animation for the NIST AM-Bench 2022 XPad multi-track benchmark simulated using the conventional single-conical heat-source model. The temperature histories correspond to the XPad multi-track validation presented in the main manuscript and illustrate the repeated thermal cycles produced by successive laser tracks. The histories are used to determine the Pad Time Above Melt (PTAM) and Pad Solid Cooling Rate (PSCR). This resource provides the temperature-history counterpart to the transient simulation presented in Online Resource 3.

Online Resource 11 (ESM_11.mp4).
Temperature-history animation for the NIST AM-Bench 2022 XPad multi-track benchmark simulated using the proposed Double Conical Heat Source (DCHS) model. The temperature histories correspond to the DCHS XPad multi-track validation presented in the main manuscript and illustrate the repeated heating and cooling cycles caused by successive laser tracks and remelting. These histories are used to determine the Pad Time Above Melt (PTAM) and Pad Solid Cooling Rate (PSCR). This resource provides the temperature-history counterpart to the transient DCHS simulation presented in Online Resource 4.

Online Resource 12 (ESM_12.mp4).
Temperature-history animation for the NIST AM-Bench 2025 single-track case simulated using the proposed Double Conical Heat Source (DCHS) model. This single-track case was used to calibrate the DCHS model prior to its application to the AM-Bench 2025 Challenge 7 multi-track prediction cases presented in the main manuscript. The animation illustrates the transient heating and cooling response at the selected monitoring location and the temperature thresholds used to evaluate the corresponding thermal metrics. This resource provides the temperature-history counterpart to the transient simulation presented in Online Resource 5.

Online Resource 13 (ESM_13.mp4).
Temperature-history animation for the NIST AM-Bench 2025 Challenge 7 multi-track case with a laser turnaround time of 5 ms, simulated using the proposed Double Conical Heat Source (DCHS) model. This case is one of the primary multi-track prediction cases of the AM-Bench 2025 competition. The animation illustrates the repeated heating, remelting, and cooling cycles at the selected monitoring locations during successive laser tracks. The temperature histories are used to evaluate the Pad Time Above Melt (PTAM) and Pad Solid Cooling Rate (PSCR) reported in the main manuscript. This resource provides the temperature-history counterpart to the transient simulation presented in Online Resource 6.

Online Resource 14 (ESM_14.mp4).
Temperature-history animation for the NIST AM-Bench 2025 Challenge 7 multi-track case with a laser turnaround time of 0.75 ms, simulated using the proposed Double Conical Heat Source (DCHS) model. This case is the second primary multi-track prediction case of the AM-Bench 2025 competition. The animation illustrates the repeated heating, remelting, and cooling cycles at the selected monitoring locations during successive laser tracks. The temperature histories are used to evaluate the Pad Time Above Melt (PTAM) and Pad Solid Cooling Rate (PSCR) reported in the main manuscript. This resource provides the temperature-history counterpart to the transient simulation presented in Online Resource 7.

S3. Supporting Numerical Data

Online Resource 15 (ESM_15.xlsx).
Grid-independence and numerical-resolution data used to assess the sensitivity of the predicted melt-pool dimensions to computational grid spacing. The file contains the grid cases and the corresponding simulation results used to justify the mesh resolution adopted in the manuscript.

Online Resource 16 (ESM_16.xlsx).
Supporting AM-Bench 2022 single-track thermal data. The workbook contains temperature-history and cooling-rate information used in the single-track calibration and validation of the thermal-CFD model.

Online Resource 17 (ESM_17.xlsx).
Supporting AM-Bench 2022 XPad data. The workbook contains numerical results used in the multi-track evaluation, including pad-related geometric and/or thermal quantities reported in the manuscript.

Online Resource 18 (ESM_18.xlsx).
Supporting AM-Bench 2025 Challenge 7 thermal data for the 5 ms and 0.75 ms turnaround-time cases. The workbook contains the numerical values used to evaluate PTAM, PSCR, and the spatial variation of the predicted thermal response.

Online Resource 19 (ESM_19.xlsx).
Supporting AM-Bench 2025 Challenge 7 dimensional data for the 5 ms and 0.75 ms turnaround-time cases. The workbook contains the simulated pad dimensions and overlap-related quantities used for comparison with the benchmark measurements.

Summary of Supplementary Files

Resource	File	Content
Online Resource 1	ESM_1.mp4	AM-Bench 2022 single track, single-conical model
Online Resource 2	ESM_2.mp4	AM-Bench 2022 single track, DCHS model
Online Resource 3	ESM_3.mp4	AM-Bench 2022 XPad, single-conical model
Online Resource 4	ESM_4.mp4	AM-Bench 2022 XPad, DCHS model
Online Resource 5	ESM_5.mp4	AM-Bench 2025 single-track case
Online Resource 6	ESM_6.mp4	AM-Bench 2025 multi-track, 5 ms turnaround
Online Resource 7	ESM_7.mp4	AM-Bench 2025 multi-track, 0.75 ms turnaround
Online Resource 8	ESM_8.mp4	Temperature history, AM-Bench 2022 single track, single-conical
Online Resource 9	ESM_9.mp4	Temperature history, AM-Bench 2022 single track, DCHS
Online Resource 10	ESM_10.mp4	Temperature history, AM-Bench 2022 XPad, single-conical
Online Resource 11	ESM_11.mp4	Temperature history, AM-Bench 2022 XPad, DCHS
Online Resource 12	ESM_12.mp4	Temperature history, AM-Bench 2025 single track
Online Resource 13	ESM_13.mp4	Temperature history, AM-Bench 2025, 5 ms turnaround
Online Resource 14	ESM_14.mp4	Temperature history, AM-Bench 2025, 0.75 ms turnaround
Online Resource 15	ESM_15.xlsx	Grid-independence / numerical-resolution results
Online Resource 16	ESM_16.xlsx	AM-Bench 2022 single-track thermal data
Online Resource 17	ESM_17.xlsx	AM-Bench 2022 XPad supporting data
Online Resource 18	ESM_18.xlsx	AM-Bench 2025 thermal data, 5 ms and 0.75 ms
Online Resource 19	ESM_19.xlsx	AM-Bench 2025 dimensional data, 5 ms and 0.75 ms

Data Availability Note

The supplementary files provide the numerical and visual material directly supporting the analyses presented in the manuscript. Additional simulation files and related data are available through the project repository and may also be obtained from the corresponding author upon reasonable request.